

Causal survival estimands under the AFT data-fusion model with HDP error

Working notes

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1 Setup

We work under the AFT decomposition

$$\log T_i = m_0(X_i, S_i) + A_i \tau(X_i) + (1 - S_i) A_i c(X_i) + \sigma \varepsilon_i, \quad \varepsilon_i \sim P, \quad (1)$$

with P assigned a hierarchical Dirichlet process (HDP) prior with Gaussian kernels and atoms shared across the two data sources. Conditional on the sampler state, the implied error CDF in source s is

$$\Phi_s(z) = \sum_{k=1}^K \pi_{s,k} \Phi^N\left(\frac{z - \mu_k}{\tau_k}\right), \quad (2)$$

where Φ^N is the standard normal CDF, $\{(\mu_k, \tau_k)\}_{k=1}^K$ are the cluster atoms (shared across sources via the top-level DP), and $\pi_{s,k}$ are the source-specific weights.

The treatment-specific AFT shift at target population s_t is

$$\eta_a(x, s_t) := m_0(x, s_t) + a \tau(x) + (1 - s_t) a c(x), \quad a \in \{0, 1\}. \quad (3)$$

The potential survival function is

$$S_a(t \mid x, s_t) = 1 - \Phi_{s_t}\left(\frac{\log t - \eta_a(x, s_t)}{\sigma}\right), \quad (4)$$

and we collect causal estimands as functionals of $\{S_0(\cdot \mid x, s_t), S_1(\cdot \mid x, s_t)\}$.

For the closed-form expressions below it is convenient to define, for each component k of the HDP mixture,

$$\xi_{a,k}(x, s_t) := \eta_a(x, s_t) + \sigma \mu_k, \quad \omega_k := \sigma \tau_k, \quad (5)$$

so that the per-component survival function under the Gaussian-kernel HDP is lognormal: $1 - \Phi^N((\log t - \xi_{a,k})/\omega_k)$.

2 The survival difference Δ_{SD}

2.1 Definition

The survival (risk) difference at time t is

$$\Delta_{SD}(t; x, s_t) := S_1(t \mid x, s_t) - S_0(t \mid x, s_t). \quad (6)$$

2.2 Closed form under the HDP

$$\Delta_{\text{SD}}(t; x, s_t) = \sum_{k=1}^K \pi_{s_t, k} \left[\Phi^{\text{N}} \left(\frac{\log t - \xi_{0,k}(x, s_t)}{\omega_k} \right) - \Phi^{\text{N}} \left(\frac{\log t - \xi_{1,k}(x, s_t)}{\omega_k} \right) \right]. \quad (7)$$

2.3 Interpretation

$\Delta_{\text{SD}}(t; x, s_t)$ is the absolute increase in the probability of surviving past time t attributable to assigning treatment versus control for individuals with covariates x in target population s_t . It lives on the probability scale, is bounded between -1 and 1 , and has a direct clinical reading: “treatment increases the chance of surviving to year t by Δ_{SD} percentage points.”

This is the absolute risk reduction (ARR), or its survival-scale analogue. It is the estimand most directly tied to clinical decision-making at a fixed horizon and is the natural quantity for computing the number-needed-to-treat $1/|\Delta_{\text{SD}}(t; x, s_t)|$ when the contrast is favourable (Cook and Sackett, 1995).

When to use. When a single clinically meaningful horizon exists — for example, “5-year survival in early-stage cancer” or “1-year survival post-transplant” — the SD is the most interpretable summary. It depends on the error distribution Φ_{s_t} through (7) and is therefore sensitive to misspecification of the tail when t extends beyond the RCT follow-up; in the present framework that sensitivity is absorbed by the HDP error.

Limitations. $\Delta_{\text{SD}}(t; x, s_t)$ is a pointwise contrast at a single t and can give a misleading picture when survival curves cross or when the gain is concentrated outside the chosen window. It does not summarise the full survival benefit. For these reasons, regulatory and HTA guidance recommends pairing the SD with an integrated summary such as the RMST (Royston and Parmar, 2013).

3 The restricted mean survival time difference Δ_{RMST}

3.1 Definition

The RMST under treatment a at horizon t^* is

$$\mu_a(t^*; x, s_t) := \int_0^{t^*} S_a(u \mid x, s_t) du, \quad (8)$$

and the RMST contrast is

$$\Delta_{\text{RMST}}(t^*; x, s_t) := \mu_1(t^*; x, s_t) - \mu_0(t^*; x, s_t). \quad (9)$$

3.2 Closed form under the HDP

The per-component lognormal RMST has the standard expression

$$\mu_a^{(k)}(t^*; x, s_t) = e^{\xi_{a,k} + \omega_k^2/2} \Phi^{\text{N}} \left(\frac{\log t^* - \xi_{a,k} - \omega_k^2}{\omega_k} \right) + t^* \left[1 - \Phi^{\text{N}} \left(\frac{\log t^* - \xi_{a,k}}{\omega_k} \right) \right], \quad (10)$$

and the contrast is the source-specific mixture

$$\Delta_{\text{RMST}}(t^*; x, s_t) = \sum_{k=1}^K \pi_{s_t, k} [\mu_1^{(k)}(t^*; x, s_t) - \mu_0^{(k)}(t^*; x, s_t)]. \quad (11)$$

3.3 Interpretation

$\Delta_{\text{RMST}}(t^*; x, s_t)$ is the expected gain in survival time within the window $[0, t^*]$ attributable to treatment, for individuals with covariates x in target population s_t . It lives on the *time scale*, in the same units as the outcome, and has a direct reading: “treatment yields Δ_{RMST} additional units of expected life within the first t^* units.”

The RMST has been advanced as a principal alternative to the hazard ratio, particularly when proportional hazards is implausible (Royston and Parmar, 2011, 2013; Uno et al., 2014). Its appeal is threefold: it has a transparent units-of-time interpretation; it is well-defined under non-proportional hazards including crossing survival curves; and it integrates the full survival benefit within the window rather than reporting a contrast at a single point. Uno et al. (2014) review its use in oncology trials and provide guidance on the choice of t^* .

When to use. The RMST is the natural headline estimand under the long-term-extrapolation goal of this work. Reporting $\Delta_{\text{RMST}}(t^*; x, s_t)$ at multiple horizons — e.g. at t^* inside the trial window, at t^* just past it, and at t^* far beyond it — traces how the survival benefit evolves with horizon and exposes the contribution of the HDP-tail to the inference.

Choice of t^* . The horizon t^* should be chosen ex ante and motivated by clinical relevance or by regulatory context. Royston and Parmar (2013) recommend t^* at or near the largest follow-up time at which a non-negligible proportion of the at-risk set remains under observation. In the present setting that recommendation is too conservative: the central methodological point is to extrapolate beyond the RCT follow-up, so t^* should deliberately exceed it.

Limitations. The RMST depends on t^* , and substantive conclusions can shift when t^* is varied. Beyond the OS follow-up, Δ_{RMST} inherits the full extrapolation uncertainty of the posterior and should be reported with appropriate credible intervals.

4 The acceleration factor and the CATE (log AF)

Going forward we focus on the acceleration factor AF and the conditional log acceleration factor τ . These are the same object on two scales — $\tau(x) = \log \text{AF}(x)$ — and they are the natural causal estimands under the AFT structural model. They are interpretable, horizon-free, and (as we will see) free of the built-in selection bias that plagues the hazard ratio.

4.1 Definition

Under the AFT model (1), the treatment-versus-control contrast at fixed covariates x is a constant additive shift on the log-time scale. Equivalently, on the time scale it is a multiplicative *acceleration*

factor. The causal acceleration factor at x is

$$\text{AF}_{\text{causal}}(x) := \exp\{\tau(x)\}, \quad \tau(x) = \log \text{AF}_{\text{causal}}(x), \quad (12)$$

i.e. the conditional log acceleration factor is the CATE on the log-time scale.

The implied relationship between potential survival functions is the quantile-shifting identity

$$S_1(t | x) = S_0(\text{AF}_{\text{causal}}(x) \cdot t | x), \quad (13)$$

so that survival under treatment at time t equals survival under control at the rescaled time $\text{AF}_{\text{causal}}(x)t$ (Cox and Oakes, 1984; Wei, 1992; Kalbfleisch and Prentice, 2002). The acceleration factor parameterises the entire contrast between the two survival curves through a single multiplicative time-scale parameter.

Causal versus observed contrast in the OS. The corresponding *observed* contrast in the OS is $\text{AF}_{\text{obs,OS}}(x) = \exp\{\tau(x) + c(x)\}$. The discrepancy $\text{AF}_{\text{obs,OS}}(x)/\text{AF}_{\text{causal}}(x) = \exp\{c(x)\}$ quantifies the multiplicative bias on the time scale due to unmeasured confounding in the OS. By transportability of potential outcomes (Assumption 4 of the framework document), $\text{AF}_{\text{causal}}(x)$ is the same in the RCT and OS populations: it is a property of the causal mechanism, not of the source.

4.2 Closed form and HDP-independence

The acceleration factor is a direct functional of the BART forest for τ and does not involve the error distribution:

$$\boxed{\text{AF}_{\text{causal}}(x) = \exp\{\tau(x)\}}. \quad (14)$$

At each MCMC iteration, $\exp\{\tau^{(r)}(x)\}$ yields a posterior draw of $\text{AF}_{\text{causal}}(x)$ without reference to the cluster atoms or weights of the HDP. Equivalently, posterior inference for $\tau(x)$ is posterior inference for the log acceleration factor. In the context of long-term extrapolation (goal G2 of the project), this means the acceleration factor is the one causal estimand that is unaffected by the HDP tail and the extrapolation uncertainty it carries, in contrast to Δ_{SD} and Δ_{RMST} .

4.3 Causal validity in the presence of unmeasured heterogeneity

The clinical and methodological appeal of the acceleration factor rests on its causal interpretation, which has recently been formalised by Brathovde et al. (2024) under a structural-causal-model formulation of the AFT. We summarise their result here because it is central to the positioning of this work.

Brathovde et al. (2024) consider structural assignments for the potential outcomes

$$T^0 := \exp\{f_0(U_0) + N_T\}, \quad T^a := T^0 / \exp\{f_1(U_1, a)\}, \quad (15)$$

where U_0 represents *frailty* (unmeasured heterogeneity in the baseline hazard), U_1 represents *effect heterogeneity* (unmeasured modification of the treatment effect), and N_T is residual noise with $N_T \perp\!\!\!\perp (U_0, U_1)$. The conditional causal acceleration factor at $(U_0, U_1) = (u_0, u_1)$ is then $\theta_c(u_0, u_1) = \exp\{f_1(u_1, a)\}$ and is invariant in u_0 : frailty does not affect the causal AF interpretation (Keiding et al., 1997; Aalen et al., 2015).

Their key identification result (Brathovde et al., 2024, Theorem 3.1) states that, under no confounding, the *observed* acceleration factor

$$\theta_m(t) := \frac{1}{t} S_{T|A=0}^{-1}(S_{T|A=a}(t)) \quad (16)$$

equals the marginal causal acceleration factor at every t in the support of the survival times. Equivalently, the AFT estimand identifies the causal contrast even when both frailty and effect heterogeneity are present in the population — a property the hazard ratio does *not* share (see also Hernán, 2010; Martinussen et al., 2020; Post et al., 2024).

Under unmeasured confounding, θ_m no longer identifies θ directly. The standard remedy — conditioning on a sufficient adjustment set L — yields a conditional observed acceleration factor that equals the conditional causal AF, and integrating over F_L recovers θ (Brathovde et al., 2024). In our setup the same logic applies to X in the RCT (Assumption 2); in the OS, the residual bias is encoded in $c(x)$, and $\text{AF}_{\text{causal}}(x) = \exp\{\tau(x)\}$ is identified from the joint RCT+OS data by the decomposition (1).

4.4 The acceleration factor versus the hazard ratio

The hazard ratio is the conventional summary in survival analysis, but it is now well-understood to be a problematic causal estimand under unmeasured heterogeneity. The literature on this point is mature:

- Hernán (2010) (“The hazards of hazard ratios”) argues that the HR is selection-biased because it conditions on survival, so that the survivor populations being compared at time $t > 0$ no longer have exchangeable distributions of unmeasured prognostic variables.
- Aalen et al. (2015) formalise this: even in a randomised trial, the HR estimated by Cox regression does not in general coincide with the causal hazard ratio when frailty is present.
- Martinussen et al. (2020) review the subtleties and give worked examples in which the HR moves over time as a direct consequence of selection on frailty.
- Post et al. (2024) prove that the Cox estimand differs from the causal effect of interest under frailty and effect heterogeneity, and that the AF maintains the causal interpretation in both cases.

The acceleration factor avoids this problem because it is defined on the survival scale (a contrast between full marginal distributions) rather than on the hazard scale (a contrast between conditional rates among survivors). It does not require conditioning on survival, and is therefore not subject to built-in selection bias (Brathovde et al., 2024). For a project whose central methodological aim is to combine an RCT with an observational study — both of which carry unmeasured heterogeneity — the acceleration factor is the most defensible causal target.

4.5 Quantile equivariance

Under the AFT model, the acceleration factor equals the ratio of survival quantiles at every level:

$$\frac{\text{Quantile}_q(T(1) | x)}{\text{Quantile}_q(T(0) | x)} = \text{AF}_{\text{causal}}(x) \quad \text{for all } q \in (0, 1), \quad (17)$$

a property called *quantile equivariance* (Wei, 1992; Hougaard, 1999). Three operational consequences:

1. *Median survival ratio.* The most clinically natural instance: the AF is the ratio of treated to control median survival times. “Treatment increases median survival by $(\text{AF}_{\text{causal}}(x) - 1) \times 100\%$ ” is a direct reading.
2. *Any quantile, same number.* The 25th-percentile ratio, the 90th-percentile ratio, and the median ratio all equal $\text{AF}_{\text{causal}}(x)$. The summary is therefore robust to the choice of survival quantile, in contrast to time-point survival differences.
3. *Direct comparability of curves.* Equation (13) states that the treated and control survival curves differ by a horizontal stretch on the log-time axis, with stretch factor $\text{AF}_{\text{causal}}(x)$. This makes visualisation of the effect straightforward: overlay the two curves on a log-time axis, and the AF is the horizontal displacement.

4.6 Conditional, marginal, and time-varying AFs under heterogeneity

In the present framework $\tau(x) = \log \text{AF}_{\text{causal}}(x)$ is a *conditional* log acceleration factor, conditional on the observed covariates $X = x$. Two subtleties about marginalising over the population deserve attention.

Time-invariance is a property of the conditional AF. Under (1), $\text{AF}_{\text{causal}}(x)$ does not depend on t : treatment is a single time-scale acceleration at each x . This is the AFT shift assumption, and it is what makes extrapolation beyond the RCT follow-up tractable.

Marginal AFs can be time-varying even when conditional AFs are not. Brathovde et al. (2024) show, however, that even when each individual has a time-invariant causal acceleration factor, the *marginal* (i.e. population-level) AF can be time-varying when there is heterogeneity in the AF across the population. This is a consequence of the marginalisation itself: at early survival times, the population contains a different mix of low-AF and high-AF individuals than at late survival times, because survival probabilities at any fixed t differ across the AF strata. The practical consequence is that a constant individual AF cannot be distinguished from a time-varying homogeneous AF on the basis of population-level data alone. Conditioning on X (and ideally on as much of the relevant heterogeneity as can be measured) is therefore central.

For our reporting strategy, this implies:

- The headline summaries should be on the *conditional* AF $\text{AF}_{\text{causal}}(x)$, e.g. via posterior projections of $\tau(x)$ onto a small basis (cf. Woody–Carvalho–Murray projection summaries in the framing document).
- Population-averaged summaries should distinguish $\mathbb{E}[\text{AF}_{\text{causal}}(X)]$ from $\exp\{\mathbb{E}[\tau(X)]\}$; the two coincide only under degenerate heterogeneity.
- Time-varying acceleration factor models (Crowther et al., 2023; Pang et al., 2021) are available if the AFT shift assumption is itself questionable; we view these as a sensitivity analysis rather than a default.

4.7 Clinical use and reporting

The acceleration factor has a direct clinical reading that requires no training in survival analysis. “ $\text{AF}_{\text{causal}}(x) = 1.5$ means that, for a patient with profile x , treatment extends time-to-event by 50% relative to control” is communicable, transparent, and on the same scale as the outcome itself. We recommend the following reporting practices:

1. Report $\text{AF}_{\text{causal}}(x)$ on the multiplicative scale (e.g., $\text{AF} = 1.5$) and the CATE $\tau(x)$ on the log scale side by side. The log scale is preferred for inference (additive, symmetric posterior); the multiplicative scale is preferred for clinical communication.
2. Report subgroup or covariate-conditional AFs via posterior projection of $\tau(x)$ onto an interpretable basis, with credible intervals derived from the projection posterior.
3. Pair AFs with at least one absolute summary — the SD at a clinically meaningful t , or $\Delta_{\text{RMST}}(t^*)$ at a clinically meaningful t^* — to communicate the *magnitude* of the absolute benefit. A 50% time-scale acceleration on a baseline median of one month and on a baseline median of ten years have very different clinical implications.
4. Where extrapolation is concerned (goal G2), prefer the AF as the headline summary at long horizons: it is the estimand least contaminated by tail uncertainty.

Caveat: AFT shift as a structural assumption. The acceleration-factor framework depends on the AFT shift structure of (1): that the treatment effect at fixed x is a single multiplicative acceleration of the time scale that does not vary with t . When this is doubtful (waning treatment effects, mode-of-failure shifts, late-emerging benefits), one can either (i) accept the AFT summary as a parsimonious approximation and discuss in sensitivity, or (ii) generalise to a time-varying acceleration factor $\text{AF}(x, t)$ via the flexible parametric approach of Crowther et al. (2023) or the time-dependent AFT extensions of Pang et al. (2021). The latter generalisation comes at the cost of giving up the single-number summary that motivates the AF in the first place.

Note on time-varying treatment regimes. For studies with time-varying treatments and treatment–confounder feedback, the structural-causal-AFT framework of Hernán et al. (2005) (using g -estimation) is required. Our setting — a single point-in-time treatment assignment — falls within the scope of Brathovde et al. (2024)’s simpler structural AFT and does not require this generalisation.

5 Population-averaged versions

Each of the three estimands has a population-averaged counterpart obtained by marginalising over the covariate distribution of the target population:

$$\Delta_{\text{SD}}(t; s_t) = \mathbb{E}[\Delta_{\text{SD}}(t; X, s_t) \mid S = s_t], \quad (18)$$

$$\Delta_{\text{RMST}}(t^*; s_t) = \mathbb{E}[\Delta_{\text{RMST}}(t^*; X, s_t) \mid S = s_t], \quad (19)$$

$$\text{AF}_{\text{causal}} = \mathbb{E}[\text{AF}_{\text{causal}}(X)] \quad \text{or} \quad \exp(\mathbb{E}[\tau(X)]), \quad (20)$$

where the last definition distinguishes the population-averaged AF (which preserves the multiplicative interpretation pointwise but is no longer a ratio of population means) from the AF of the

population-averaged log-time shift (which is). The two coincide only under degenerate $\tau(X)$. Point-wise reporting via the posterior projection summaries is generally preferable to a single marginal number unless heterogeneity is small.

The Monte Carlo estimator is in each case the empirical average over evaluation rows of the corresponding posterior draw, with Bayesian-bootstrap weights if covariate-distribution uncertainty is to be propagated.

6 Summary

- $\Delta_{\text{SD}}(t; x, s_t)$ — absolute survival-probability gain at fixed t . Probability scale; depends on HDP tail. Natural for fixed clinical horizons and number-needed-to-treat.
- $\Delta_{\text{RMST}}(t^*; x, s_t)$ — expected life-time gain within $[0, t^*]$. Time scale; depends on HDP tail. Natural headline estimand under the long-term extrapolation goal.
- $\text{AF}_{\text{causal}}(x) = \exp\{\tau(x)\}$ — multiplicative time-scale acceleration. Horizon-free; HDP-free; assumes the AFT shift structure. Natural complement summarising treatment heterogeneity in a single covariate-specific number.

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